Case Study: Creating and Managing an IT Helpdesk Ticket System in MySQL (CMD)

Issue Summary:

To simulate a real-world IT environment, I created a simple ticketing system using MySQL from the Command Line Interface (CMD). This demonstrates skills in database creation, table design, data entry, and query execution — all critical for IT support professionals who interact with backend tools or assist developers/sysadmins.

Tools Used:

- MySQL Server (installed om Windows VM)
- Command Prompt (CMD)
- Test scenario: "IT Ticketing Table"

Steps Taken:

1. Log into MySQL.
2. Show databases.
3. Create new database.
4. Use that database.
5. Create table tickets.
6. Insert records.
7. View all records.
8. Update ticket.
9. Delete ticket.
10. Exit MySQL.

Result:

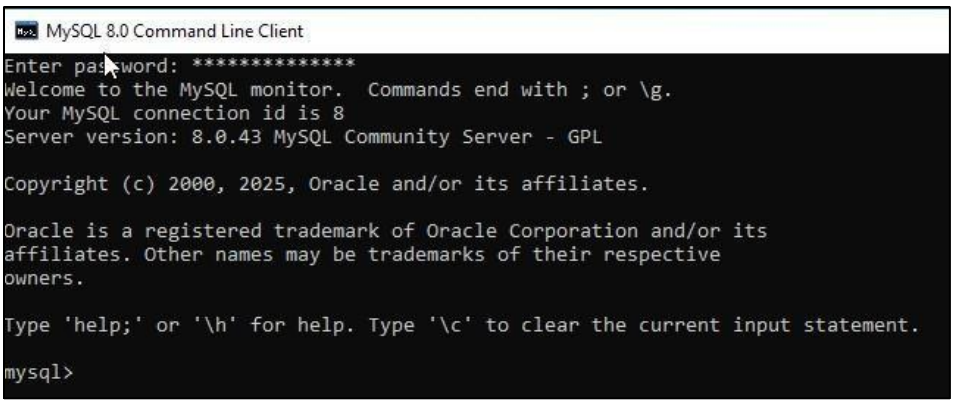
- Successfully created and managed a simple IT ticketing system using MySQL and CMD.
- Demonstrated full lifecycle of a record: create, read, update, delete (CRUD).
- Gained hands-on experience with SQL queries and database structure.

What I Learned:

- How to interact with MySQL through command line.
- How to define tables, insert, and query data.
- The basics of SQL data types and auto-incrementing IDs.
- How updates and deletions work in relational databases.

Screenshots:

- 01_mysql_login.png



```
MySQL 8.0 Command Line Client
Enter password: **************
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.43 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```
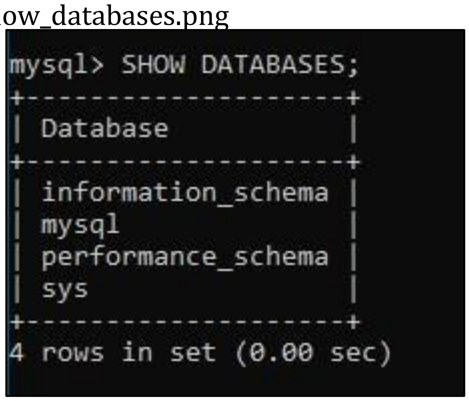

- 02_show_databases.png



```
mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
4 rows in set (0.00 sec)
```

-03_create_database.png

```
mysql> CREATE DATABASE itsupport;
Query OK, 1 row affected (1.80 sec)

mysql>
```

-04_use_database.png

```
mysql> USE itsupport;
Database changed
mysql>
```

-05_create_table.png

```
mysql> CREATE TABLE tickets (
    -> id INT AUTO_INCREMENT PRIMARY KEY,
    -> username VARCHAR(100),
    -> issue VARCHAR(255),
    -> status VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.05 sec)
```

-06_insert_data.png

```
mysql> INSERT INTO tickets (username, issue, status)
    -> VALUES
    -> ('jane.doe', 'Printer not responding', 'open'),
    -> ('john.doe', 'PC not booting', 'open'),
    -> ('Alex.hudson', 'WI-FI not connecting', 'open');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

-07_select_all.png

```
mysql> SELECT * FROM  tickets;
+----+-------------+-------------------------+--------+
| id | username    | issue                   | status |
+----+-------------+-------------------------+--------+
|  1 | jane.doe    | Printer not responding  | open   |
|  2 | john.doe    | PC not booting          | open   |
|  3 | Alex.hudson | WI-FI not connecting    | open   |
+----+-------------+-------------------------+--------+
3 rows in set (0.00 sec)
```

-

08_update_ticket.png

```
mysql> UPDATE tickets SET status='closed' WHERE id=1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

-09_delete_ticket.png

```
mysql> DELETE FROM tickets WHERE id=2;
Query OK, 1 row affected (0.01 sec)
```

-10_updated_table.png

```
mysql> SELECT * FROM tickets;
+----+-------------+-------------------------+--------+
| id | username    | issue                   | status |
+----+-------------+-------------------------+--------+
|  1 | jane.doe    | Printer not responding  | closed |
|  3 | Alex.hudson | WI-FI not connecting    | open   |
+----+-------------+-------------------------+--------+
2 rows in set (0.00 sec)
```